



Chathurya V R 251801045-2025-AIDS ▾

C2**Started on** Thursday, 27 August 2026, 8:52 AM**State** Finished**Completed on** Thursday, 17 September 2026, 9:41 AM**Time taken** 21 days**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  // Recursive Binary Search function to find a target value in a sorted array range [low, high]
4  int binarySearch(int arr[], int low, int high, int target) {
5      if (low > high) {
6          return -1; // Not found
7      }
8
9      int mid = low + (high - low) / 2;
10
11     if (arr[mid] == target) {
12         return mid; // Found target index
13     } else if (arr[mid] > target) {
14         return binarySearch(arr, low, mid - 1, target); // Search left half
15     } else {
16         return binarySearch(arr, mid + 1, high, target); // Search right half
17     }
18 }
19
20 // Divide and Conquer wrapper function to check for the two elements
21 int findTwoElements(int arr[], int low, int high, int x, int *elem1, int *elem2) {
22     // Base case: if low crosses high, no pair found in this branch
23     if (low >= high) {
24         return 0;
25     }
26
27     // Divide: Pick the current element as the first candidate
28     int i = low;
29     int target = x - arr[i];
30
31     // Conquer: Use recursive binary search to find the complement in the rest of the array [i + 1, high]
32     int j = binarySearch(arr, i + 1, high, target);
33
34     if (j != -1 && i != j) {
35         *elem1 = arr[i];
36         *elem2 = arr[j];
37         return 1; // Pair found
38     }
39
40     // Otherwise, recursively check the rest of the array (Divide and Conquer step)
41     return findTwoElements(arr, low + 1, high, x, elem1, elem2);
42 }
43
44 int main() {
45     int n;
46     scanf("%d", &n);
47
48     int arr[n];
49     for (int i = 0; i < n; i++) {
50         scanf("%d", &arr[i]);
51     }
52

```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)